Task 3 – Due April 12 at 12:00 (noon)

I would like you to have a closer look at Artificial Intelligence ETFs. Artificial Intelligence ETFs are funds that meet at least one of the following three criteria:

- They are funds that specifically invest in companies involved in the development of new products or services, technological improvements in scientific research related to artificial intelligence, or
- They are funds that have at least 25% of portfolio exposure to companies that spend large amounts on artificial intelligence research and development (R&D) expenses. Examples of such companies are Amazon, Tesla Motors, Apple and Alphabet, or
- They are funds that use artificial intelligence methodologies to select individual securities for inclusion into the fund.

Main aim of your analysis is to explore time series properties of these ETFs and how (a portfolio constructed from) these ETFs performs in terms of risk/return and produce forecasts for future performance.

- Global X Robotics & Artificial Intelligence ETF (BOTZ)
- WisdomTree Artificial Intelligence and Innovation Fund (WTAI)
- Invesco AI and Next Gen Software ETF (IGPT)
- ROBO Global Robotics & Automation Index ETF (ROBO)
- Global X Artificial Intelligence & Technology ETF (AIQ)

Obtain price information for 2023, 2024 and 2025 (three full years of daily data) for these AI ETFs. ETFs data is available from yahoo finance (and I showed you the code example of how to get the data in Tutorials).

1. Plot the five series in one plot for one of the price points (open, close, high, low) (Hint: you may need to ensure your dates are read by R as dates). Clearly indicate your chosen price point. Make sure axes, legend, etc. is tidy. Comment briefly what you observe.

2. Pick one ETF series and one price point (open/close/high/low) and make it a time series. (For simplicity, if you struggle, you can just say the time series are daily, ignoring that the values are observed only on working days). Decompose the time series of your choice using decompose function. Is this time series stationary? Why? Perform Augmented Dicky-Fuller test for unit root. Interpret the test. If your series is not stationary, can you make your time series stationary? Try some transformation. Did you get to stationarity?

3. If/once your series is stationary, what is your stationary process? (White noise? AR(p)? MA(q)?) Plot ACF and PACF. Interpret what you see.

Calculate returns (at your choice of data frequency and your choice of method to calculate the returns) for all assets (5 AI ETFs).

4. Plot return series in one graph and report main summary statistics for each of the return series. Clearly indicate what parameters you chose to calculate returns. Make sure axes, legend, etc. is tidy. Comment briefly what you observe. What would be an ETF of your choice to invest in? Why?

5. Construct an equally weighted portfolio from AI ETFs (so each ETF has a portfolio weight of 1/5) and report portfolio volatility and returns (plot and report main characteristics).

6. Construct an alternative portfolio (feel free to use weights of your choice) from the AI ETFs. Again, report portfolio characteristics.

7. For each of the two portfolios (equal weighted and with weights of your choice) calculate their CAPM betas (feel free to use S&P 500 (^GSPC) as your benchmark for the market). What do you see? Interpret your findings briefly (2-3 sentences).

8. Run a Monte Carlo (MC) simulation for the performance (in terms of daily return) of one of the AI ETFs for the next 30 days. Clearly state assumptions employed in your simulation. Interpret simulation results (plot!).

9. Run a Monte Carlo (MC) simulation for the performance (in terms of daily return) of any of the portfolios analyzed above for the next 30 days. Clearly state assumptions employed in your simulation. Interpret simulation results (plot!).

10. Perform a sensitivity analysis to your MC simulation done in part 9: perform more simulations under different simulation assumptions (hint: change seed a couple of times!) and simulate for different forecasting periods (hint: try simulating for 60 days and compare the result to the one from 30 day simulation). Discuss your findings.

All parts carry equal weight.

Hand in your answer to me via e-mail to nora.laurinaityte@ilte.lt or submit via TEAMS. Please include all the commands used in your work in addition to the plots etc. that have been requested, i.e. submit a code (ideally copied into MS Word file) to produce all the results (comment to answer questions in the Task directly in your code that produces calculations and graphs from raw data; otherwise submit an answer sheet where you include your code) as well as the results themselves.